

1. Research Interests

Artificial intelligence and its applications to scientific discoveries, quantum AI, quantum circuit compilation and optimization for near-term and fault-tolerance devices, Hamiltonian simulation, classical simulation of quantum and chemical systems, mathematical methods for quantum computing

2. Education

African Institute for Mathematical Sciences (AIMS), South Africa September 2025 – July 2026
Master of Science in Mathematical Sciences with specialization in Artificial Intelligence for Science.

- **Final Grade:** Distinction | **Thesis:** [Generative Hamiltonian Simulation](#)
- **Supervisor:** Dr. Ryan Sweke | **Examiner:** Dr. Kouhei Nakaji

University of Ibadan, Nigeria April 2017 – June 2023
Bachelor of Science in Mathematics

- **Final Grade:** First-Class Honours — CGPA: 3.66/4.0 | **Thesis:** [Some Applications of Group Theory to Polynomial Equations](#) | **Supervisor:** Professor Praise Adeyemo

3. Publications

- Daniel, O.A. (2026). *Generative Hamiltonian Simulation*. Master's Thesis, African Institute for Mathematical Sciences (AIMS), South Africa. DOI: [10.13140/RG.2.2.21841.08804](https://doi.org/10.13140/RG.2.2.21841.08804)
- Daniel, O. A. (2026). *Weaving the Fault-Tolerant Future*. [Medium](#)
- Daniel, O. A. (2025). *Partial Symmetry Enforced Attention Decomposition (PSEAD): A group-theoretic framework for equivariant transformers in biological systems*. [arXiv:2507.14908](https://arxiv.org/abs/2507.14908)
- Daniel, O. A. (2025). *A Comprehensive Computational Approach for Detecting Multilingual, Multicultural, and Multievent Online Polarization in English and Swahili*. DOI: [10.13140/RG.2.2.11612.35205](https://doi.org/10.13140/RG.2.2.11612.35205)
- Daniel, O. A. (2025). *Architecting a Scalable Recommender System: From Alternating Least Squares to Production-Ready API*. DOI: [10.13140/RG.2.2.11106.08649](https://doi.org/10.13140/RG.2.2.11106.08649)

4. Research Experience

MIORPA, University of Oxford July 2026 – September 2026

- Conducting exploratory research with Dr Duncan Laurie to investigate how quantum affine algebras and their representations describe the braiding of non-Abelian anyons. Learning the connections between quantum algebra to quantum gates and vice versa.
- Studying the construction of finite-dimensional representations of $U_q(\widehat{\mathfrak{sl}}_2)$ and their associated R -matrices, and investigating how the Yang–Baxter equation gives rise to representations of the braid group.
- Investigating how these braid-group representations encode quantum gates through anyon braiding, with particular focus on the Fibonacci and Ising anyon models, computational universality, and the conditions under which braiding alone can approximate arbitrary quantum operations.
- Addressing the computational and mathematical problem of obtaining explicit R -matrices and braid representations from quantum affine algebras, and examining their implications for topological quantum gate synthesis and fault-tolerant quantum computation.
- Independently studied Quantum error correction codes including GKP and cat codes, and filtered randomized benchmarking for bosonic quantum systems.

AIMS Quantum research group, AIMS, South Africa November 2025 – July 2026

- Developed a state-independent generative framework based on the Generative Quantum Eigensolver (GQE) for Hamiltonian simulation.
- Demonstrated more than a fivefold reduction in circuit depth relative to first-order Trotter decomposition while maintaining equal or higher fidelity.
- Validated the framework on the transverse-field Ising model, Heisenberg spin chains, and one- and two-dimensional Hubbard models.
- Implemented algorithms using Qiskit, combining quantum algorithms, machine learning, and numerical simulations.

Undergraduate Researcher, University of Ibadan, Nigeria December 2022 – July 2023

- Studied applications of finite group theory, Galois theory, and abstract algebra to the solvability of polynomial

equations.

- Developed a strong mathematical foundation in algebra, representation theory, and symmetry, which now underpins my research in quantum information and computational quantum physics.

Research Assistant, Lagos State University, Nigeria

November 2022 – May 2023

- Assisted in data collection and statistical analysis for research on partner support for exclusive breastfeeding in Ibadan North East LGA, Oyo State.

Research Assistant, University of Ibadan, Nigeria

November 2021 – May 2022

- Contributed to research on awareness and utilization of preconceptional folic acid and iron supplementation among women in Lagos State through data collection, analysis, and manuscript preparation.

5. Teaching Experience

Assistant Instructor, Data Science and Machine Learning, Lamzytech

April - October 2024

- Assisted in teaching Python, machine learning, and data science concepts to **5** students through lectures, hands-on coding sessions, and project supervision.
- Mentored interns on practical machine learning projects, resulting in **3** successfully completing the programme and deploying end-to-end applications.

Undergraduate Tutorial Teacher, University of Ibadan

September 2021 – July 2023

- Taught tutorial sessions in **Operations Research (MAT 353)** to class of **35** students.
- Prepared problem-solving sessions and revision classes that contributed to improved student understanding and examination performance by more than **20%** compared to the previous academic session

Group Leader, Peer Assisted Study Session, University of Ibadan

2018/2019 Academic session

- Led weekly peer-assisted learning sessions in **Calculus and Trigonometry (MAT 121)** for approximately **15** first-year students.
- Facilitated collaborative problem-solving sessions that helped improve student engagement and course grades resulting in distinction for 9 out of 15 students involved

6. Work Experience

Machine Learning Engineer, Lamzytech (Remote)

November 2023 – March 2026

- Designed, developed, and deployed two end-to-end machine learning and software solutions, integrating frontend, backend, databases, and AI models for production applications.
- Built three scalable web and mobile applications using Python, Java, TypeScript, React, React Native, Spring Boot, FastAPI, and RESTful APIs, improving service delivery by over 40%.
- Developed and optimized two machine learning pipelines using PyTorch, TensorFlow, Scikit-learn, and MLflow for model training, evaluation, deployment, and monitoring.
- Implemented four cloud-based applications and CI/CD workflows using Git, GitHub, Docker, Google Cloud Platform, and Firebase.
- Applied software engineering best practices including version control, object-oriented programming, API development, database design, performance optimization, and automated testing.

7. Service

Volunteering

- **Volunteer for the first AIMS Workshop and School on the Theory of Quantum Learning Algorithms** October 2025
 - Assisted with participant registration, workshop logistics, and coordination of scientific sessions, helping facilitate interactions between participants and invited speakers.
- **Student Intern, Lagos Bureau of Statistics / Volunteer Data Engineer, Lagos State WASH Survey (Hybrid)** October 2020 - May 2022
 - Supported data collection, cleaning, validation, and analysis for state-wide household surveys, contributing to high-quality datasets used for policy and statistical reporting.

Leadership and Committee Roles

- **Financial Secretary, Financial Inclusion Community Development Service (CDS), National Youth Service Corps (NYSC)** August - October 2024
 - Managed the finances of the CDS group, maintained transparent financial records, prepared financial reports, and coordinated budgeting for community development activities.
- **Member of Students Academic Committee, Department of Mathematics, Faculty of Science, University of Ibadan** 2021/2022 and 2022/2023 Academic sessions
 - Represented students in academic matters, contributed to discussions on curriculum and student welfare, and participated in organizing departmental academic programmes and seminars.

8. Awards and Honors

- (2025) Google DeepMind Scholarship – AI for Science, AIMS South Africa
- (2024) Excellent Leadership Award – NYSC Jos-South LGA, Plateau State
- (2019, 2021, 2023) Dean's Roll of Honor Award, Faculty of Science, University of Ibadan
- (2023) Award of Academic Excellence – Department of Mathematics, University of Ibadan

9. Talks and Seminars

- (2025) Participant – 1st AIMS Workshop and School on The Theory of Quantum Learning Algorithms. (AIMS South Africa)
- (2025) Participant – Online Senior Learning Seminar in Algebraic Geometry (Prof. Balázs Szendrői)

10. Certifications

- Qiskit Global Summer School 2026 (In progress)
- Google Quantum AI (Coursera) – Hands-on Quantum Error Correction with Google Quantum AI
- University of Michigan (Coursera) – Python for Everybody Specialization
- DeepLearning.AI (Coursera) – Deep Learning Specialization, TensorFlow Developer Specialization
- University of California (Coursera) – Initiating and Planning Projects

11. Proficiencies

- **Quantum Computing:** Qiskit, IBM Quantum Platform, Qiskit Runtime, Stim
- **Programming:** Python, Java, TypeScript, MATLAB, R, CUDA C++, Git, REST APIs, Spring Boot, React, React Native, SQL, LaTeX.
- **Professional Skills:** Scientific communication, technical presentation, analytical problem solving, project management, teamwork, leadership, software documentation.
- **Languages:** English–C1, Yoruba–Native speaker.

12. AI Projects

- **Text Analytics and Linguistic Games Suite** — Implemented a text analysis and interactive Scrabble-style word game application using Python and Gradio. The system performs sentiment analysis, readability scoring, and token frequency statistics, while providing an educational language-based gaming experience. Live Demo: [Text Analyzer](#)
- More on my [website](#)

13. References (*please notify me before contacting them*)

- AIMS Academic Director for AI for Science: Profesor Ulrich Paquet | ulrich@aims.ac.za
- Thesis Supervisor: Dr Ryan Sweke | rsweke@aims.ac.za
- Google DeepMind Mentor: Dr Kathryn Tunyasuvunakool | ktkool@deepmind.com